Folate: Fact Sheet for Consumers (original version)

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What is folate and what does it do?

Folate is a B-vitamin that is naturally present in many foods. Your body needs folate to make DNA and other genetic material. Your body also needs folate for your cells to divide. A form of folate, called folic acid, is used in fortified foods and most dietary supplements.

How much folate do I need?

The amount of folate you need depends on your age. Average daily recommended amounts are listed below in micrograms (mcg) of dietary folate equivalents (DFEs).

Life Stage	Recommended Amount
Birth to 6 months	65 mcg DFE
Infants 7–12 months	80 mcg DFE
Children 1–3 years	150 mcg DFE
Children 4–8 years	200 mcg DFE
Children 9–13 years	300 mcg DFE
Teens 14–18 years	400 mcg DFE
Adults 19+ years	400 mcg DFE
Pregnant teens and women	600 mcg DFE
Breastfeeding teens and women	500 mcg DFE

The measure of mcg DFE is used because your body absorbs more folic acid from fortified foods and dietary supplements than folate found naturally in foods. Compared to folate found naturally in foods, you actually need less folic acid to get recommended amounts. For example, 240 mcg of folic acid and 400 mcg of folate are both equal to 400 mcg DFE.

All women and teen girls who could become pregnant should consume 400 mcg of folic acid daily from supplements, fortified foods, or both in addition to the folate they get from following a healthy eating pattern.

What foods provide folate?

Folate is naturally present in many foods, and folic acid is added to some foods. You can get recommended amounts by eating a variety of foods, including the following.

Folate is naturally present in:

- Beef liver
- Vegetables (especially asparagus, brussels sprouts, and dark green leafy vegetables such as spinach and mustard greens)
- Fruits and fruit juices (especially oranges and orange juice)
- Nuts, beans, and peas (such as peanuts, black-eyed peas, and kidney beans)

Folic acid is added to the following foods:

- Enriched bread, flour, cornmeal, pasta, and rice
- Fortified breakfast cereals
- Fortified corn masa flour (used to make corn tortillas and tamales, for example)

To find out whether a food has added folic acid, look for folic acid on its Nutrition Facts label.

What kinds of folate dietary supplements are available?

Folate is available in multivitamins and prenatal vitamins. It is also available in B-complex dietary supplements and supplements containing only folate. In dietary supplements, folate is usually in the form of folic acid, but methylfolate (5-MTHF) is also used. Dietary supplements containing 5-MTHF might be better than folic acid for some individuals who have a gene variant called MTHFR C677T because their bodies can use this form more easily. However, all women and teen girls who could become pregnant should get 400 mcg a day of folic acid, not 5-MTHF, even if they have an MTHFR C677T gene variant (see Neural tube defects below).

Am I getting enough folate?

Most people in the United States get enough folate. However, certain people are more likely than others to have trouble getting enough folate:

- Teen girls age 14–18 years, women age 19–30 years, and non-Hispanic black women
- People with alcohol use disorder
- People with disorders that lower nutrient absorption (such as celiac disease and inflammatory bowel disease)
- People with an MTHFR gene variant

What happens if I don’t get enough folate?

Folate deficiency is rare in the United States, but some people do not get enough. Getting too little folate can result in megaloblastic anemia, a blood disorder that causes weakness, fatigue, trouble concentrating, irritability, headache, heart palpitations, and shortness of breath. Folate deficiency can also cause open sores on the tongue and inside the mouth as well as changes in the color of the skin, hair, or fingernails.

Women who don't get enough folate are at risk of having babies with neural tube defects, such as spina bifida. Folate deficiency can also increase the likelihood of having a premature or low birth weight baby.

What are some effects of folate on health?

Scientists are studying folate to understand how it affects health. Here are several examples of what this research has shown.

Neural tube defects

Taking folic acid before becoming pregnant and during early pregnancy helps prevent neural tube defects in babies. Neural tube defects are major birth defects in a baby's brain (anencephaly) or spine (spina bifida). However, about half of all pregnancies are unplanned. Therefore, all women and teen girls who could become pregnant should consume 400 mcg of folic acid daily from supplements, fortified foods, or both—even if they have an MTHFR C677T gene variant—in addition to the folate they get from following a healthy eating pattern.

Since 1998, the U.S. Food and Drug Administration (FDA) has required food companies to add folic acid to enriched bread, flour, cornmeal, pasta, rice, and other grain products sold in the United States. In 2016, FDA allowed manufacturers to voluntarily add folic acid to corn masa flour. Because most people in the United States eat these foods, folic acid intakes have increased since 1998, and the number of babies born with neural tube defects has decreased.

Cancer

Folate that is naturally present in food may decrease the risk of several forms of cancer, but folate supplements might have different effects on cancer risk depending on how much the person takes and when. People who take recommended amounts of folic acid before cancer develops might decrease cancer risk, but taking high doses after cancer (especially colorectal cancer) begins might speed up its progression. For this reason, people should be cautious about taking high doses of folic acid supplements (more than the upper limit of 1,000 mcg), especially if they have a history of colorectal adenomas (which sometimes turn into cancer). More research is needed to understand the roles of dietary folate and folic acid supplements in cancer risk.

Depression

People with low blood levels of folate might be more likely to have depression. In addition, they might not respond as well to antidepressant treatment as people with normal folate levels.

Folate supplements, particularly those that contain 5-MTHF, might make antidepressant medications more effective. However, whether supplements help both people with normal folate levels and those with folate deficiency isn't clear. More research is needed to better understand the role of folate in depression and whether folate supplements are helpful when used in combination with standard treatment.

Heart disease and stroke

Folic acid supplements lower levels of homocysteine, an amino acid in the blood that's linked to a higher risk of cardiovascular disease, but the supplements don't directly decrease the risk of heart disease. Some studies have shown that a combination of folic acid with other B-vitamins, however, helps prevent stroke.

Dementia, cognitive function, and Alzheimer's disease

Folic acid supplements, with or without other B-vitamins, do not seem to improve cognitive function or prevent dementia or Alzheimer's disease. However, more research on these topics is needed.

Preterm birth, congenital heart defects, and other birth defects

Taking folic acid might reduce the risk of having a premature baby or a baby with birth defects, such as certain types of heart problems. However, more research is needed to understand how folic acid affects the risk of these conditions.

Autism spectrum disorder

Autism spectrum disorder (ASD) affects communication and behavior, usually beginning by age 2. People with ASD have limited interests, repetitive behaviors, and difficulty communicating and interacting with others.

Some studies have shown that taking recommended amounts of folic acid before and during early pregnancy may help reduce the risk of ASD in the child. However, because the study results are inconclusive, more research is needed to understand the potential role of folic acid in lowering the risk of ASD.

Can folate be harmful?

Folate that is naturally present in food and beverages is not harmful. However, you should not consume folate in supplements or fortified foods and beverages in amounts above the upper limit, unless recommended by a health care provider.

Taking large amounts of folate supplements might hide a vitamin B12 deficiency because these supplements can correct the anemia that the vitamin B12 deficiency causes but not the nerve damage that the vitamin B12 deficiency also causes. The vitamin B12 deficiency can lead to permanent damage of the brain, spinal cord, and nerves. Large doses of folate supplements might also worsen the symptoms of vitamin B12 deficiency.

High doses of folic acid might increase the risk of colorectal cancer and possibly other cancers in some people. High doses can also lead to more folic acid in the body than it can use, but whether these increased folic acid levels are harmful is not completely clear.

The daily upper limits for folate from supplements and fortified foods and beverages are listed below.

Ages	Upper Limit
Birth to 6 months	Not established
Infants 7–12 months	Not established
Children 1–3 years	300 mcg
Children 4–8 years	400 mcg
Children 9–13 years	600 mcg
Teens 14–18 years	800 mcg
Adults 19+ years	1,000 mcg

Does folate interact with medications or other dietary supplements?

Folate supplements can interact with several medications. Here are some examples:

- Folate supplements could interfere with methotrexate (Rheumatrex, Trexall) when taken to treat cancer.
- Taking antiepileptic or antiseizure medications, such as phenytoin (Dilantin), carbamazepine (Carbatrol, Tegretol, Equetro, Epitol) and valproate (Depacon), could reduce blood levels of folate. Also, taking folate supplements could reduce blood levels of these medications.
- Taking sulfasalazine (Azulfidine) for ulcerative colitis could reduce the body’s ability to absorb folate and cause folate deficiency.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and medicines you take. They can tell you if those dietary supplements might interact or interfere with your prescription or over-the-counter medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients.

Folate and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the *Dietary Guidelines for Americans*.

Where can I find out more about folate?

- For general information on folate
  - ODS Health Professional Fact Sheet on Folate
- For more information on food sources of folate
  - USDA's FoodData Central
  - Nutrient List for folate (listed by food or by folate content), USDA
- For more advice on choosing dietary supplements
  - ODS Frequently Asked Questions: Which brand(s) of dietary supplements should I purchase?
- For information about building a healthy dietary pattern
  - Dietary Guidelines for Americans

Glossary

absorption
: In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

adenoma
: A type of tumor that is benign (not cancer).

Alzheimer's disease
: A brain disease in which thinking, memory, and reasoning ability is slowly destroyed. In advanced stages, an affected person becomes disoriented and confused, has mood and behavior changes, and has difficulty talking, walking, and swallowing. Alzheimer's disease is progressive, irreversible, and incurable.

amino acid
: A chemical building block of protein.

anemia
: A condition in which the number of red blood cells in the blood, or the amount of hemoglobin in them, is lower than normal, causing a condition in which red blood cells are not able to supply enough oxygen to all the tissues in the body. Hemoglobin is the substance in red blood cells that carries oxygen to your body's cells.

anencephaly
: A condition in which a baby is born without most of a brain and skull. The brain may not be covered by bone or skin. Babies born with this condition do not survive more than a few hours or days. Anencephaly belongs to the group of disorders called neural tube defects.

cardiovascular disease
: CVD. A general term referring to disorders of the heart and blood vessels. CVD includes coronary artery disease, heart failure, atherosclerosis, high blood pressure, peripheral artery disease, and stroke.

celiac disease
: An autoimmune disorder in which eating gluten (a protein found in wheat, rye, barley, and possibly oats) causes the immune system to damage the small intestine, making it unable to absorb nutrients. It is a genetic disease that sometimes becomes active for the first time after surgery, pregnancy, childbirth, viral infection, or extreme stress. Also called sprue.

cell
: The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

cognitive function
: Mental awareness and judgment.

colorectal cancer

Uncontrolled growth of abnormal cells in the colon (the longest part of the large intestine) and/or the rectum (the last several inches of the large intestine before the anus).

consume
: To eat or drink.

deficiency
: An amount that is not enough; a shortage.

deoxyribonucleic acid
: DNA. The molecules inside cells that carry genetic information and pass it from one generation to the next.

dietary fiber
: A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Folate Equivalent
: DFE. A term used to describe the Recommended Dietary Allowance of folate. DFE accounts for the easier absorption of folic acid in supplements and fortified foods compared with folate found naturally in foods, which is absorbed only about half as well. One DFE = 1 microgram (mcg) food folate = 0.6 mcg folic acid from supplements and fortified foods.

Dietary Guidelines for Americans
: Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement
: A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder
: In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

dose
: The amount of medicine or other substance taken at one time or over a specific period of time.

enriched
: When certain nutrients (such as vitamins and minerals) have been added to a food product to replace nutrients that may be lost during processing or storage. For example, white flour is enriched with thiamin, riboflavin, niacin, and iron because those nutrients are lost when grain is made into flour. Folic acid is also added to enriched flour.

fatigue
: Extreme tiredness and an inability to function due to lack of energy.

folate
: A general term for the various forms of folic acid, a B vitamin. Folate is needed to make DNA, RNA, and amino acids. It occurs naturally in foods and is found in leafy green vegetables (such as spinach and turnip greens), fruits (such as citrus fruits and juices), and dried beans and peas. The synthetic (manufactured) form of folate used in supplements and fortified foods is called folic acid.

folic acid
: The form of folate (a B vitamin occurring naturally in food) that is manufactured and used in supplements and fortified foods.

Food and Drug Administration
: FDA, Department of Health and Human Services. FDA is the Federal government agency responsible for ensuring that foods and dietary supplements are safe, wholesome and sanitary, and that drugs, medical devices, cosmetics, and food are honestly, accurately and informatively represented to the public. FDA regulates dietary supplements under a different set of regulations than those covering conventional foods and drug products (prescription and over-the-counter). The dietary supplement manufacturer is responsible for ensuring that a dietary supplement is safe before it is marketed. FDA is responsible for taking action against any unsafe dietary supplement product after it reaches the market.

Generally, manufacturers do not need to get FDA approval before producing or selling dietary supplements.

fortified
: When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gene
: The functional and physical unit of heredity passed from parent to offspring. Genes are pieces of DNA, and most genes contain the information for making a specific protein.

heart palpitation
: Forceful and irregular beating of the heart.

homocysteine
: An amino acid (a building block of protein). At high blood levels, it may increase the risk of developing coronary heart disease, heart attack, stroke, and Alzheimer's disease. Elevated homocysteine may also increase the risk of developing osteoporosis and bone fractures.

infant
: A child younger than 12 months old.

inflammatory bowel disease
: IBD. Long-lasting (chronic) problems that cause irritation and ulcers in the digestive tract. The most common disorders are ulcerative colitis and Crohn's disease.

interaction
: A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

kidney
: One of two organs that remove waste from the blood (as urine). The kidneys also make erythropoietin (a substance that stimulates red blood cell production) and help regulate blood pressure. The kidneys are located near the back under the lower ribs.

label
: When referring to dietary supplements, information that appears on the product container, including a descriptive name of the product stating that it is a "supplement"; the name and place of business of the manufacturer, packer, or distributor; a complete list of ingredients; and each dietary ingredient contained in the product. Supplements must also include directions for use, nutrition labeling in the form of a Supplement Facts panel that identifies each dietary ingredient contained in the product and the serving size, amount, and active ingredients.

liver
: A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

low birth weight
: A baby weighing less than 5.5 pounds at birth. Low birth weight babies are at risk of sudden infant death syndrome (SIDS), infections, delayed development (for example, sitting, crawling, and talking), learning disabilities, and other health conditions, such as breathing problems, cerebral palsy, and heart disorders.

medical history
: Information about a person’s health, such as allergies, illnesses, surgeries, medications, immunizations, and the results of tests and physical exams. It may also include information about health habits, such as diet and exercise, and health information about current and past illnesses of one’s parents and other close family members.

megaloblastic anemia
: A disorder in which red blood cells are larger than normal, immature, and few in number, which reduces the amount of oxygen that can be carried by the blood to the body's tissues. It is caused by a deficiency in folate or vitamin B_{12}.

methotrexate
: A drug that blocks the body's ability to use folic acid, which is needed by growing cells such as those making up the skin, blood, digestive tract, and the cells that protect the body against infection and disease. Methotrexate is used to treat some types of cancer, arthritis, and severe skin disorders. It belongs to the group of drugs called antimetabolites.

microgram
: μg or mcg. A unit of weight in the metric system equal to one millionth of a gram. (A gram is approximately one-thirtieth of an ounce.)

mineral
: In nutrition, an inorganic substance found in the earth that is required to maintain health.

multivitamin/mineral dietary supplement
: MVM. A product that is meant to supplement the diet. MVMs contain a variety of vitamins and minerals. The number and amounts of these nutrients can vary substantially by product.

nerve
: A bundle of microscopic fibers that carries messages back and forth from the brain to other parts of the body.

neural tube defects
: Birth defects of the brain and spinal cord. They happen when the neural tube, which develops into the brain and spinal cord during the first few weeks of pregnancy, doesn't close completely. One common neural tube defect is spina bifida, in which part of the spinal cord pokes through the spine. Another common neural tube defect is anencephaly, in which major parts of the brain are missing, and the baby dies.

neuron
: A nerve cell. Neurons send chemical and electrical messages throughout the nervous system that direct the body to function, move, think, and have emotions.

nutrient
: A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

nutrition
: The process of eating, digesting, and absorbing nutrients (such as protein, carbohydrate, fat, vitamins, minerals, and water) from food to maintain the body, grow new cells, repair tissues, and supply energy. Nutrition is also the science of food, diet, and health.

pharmacist
: A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

prenatal
: Before birth; during pregnancy.

prescription
: A written order from a health care provider for medicine, therapy, or tests.

prevent
: To stop from happening.

progression
: In medicine, the course of a disease as it becomes worse. For example, as cancer progresses, it spreads in the body.

risk
: The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

spina bifida
: A disorder in which a fetus's spine does not close properly during the first month of pregnancy. It may result in permanent damage to the nerves and spinal cord, causing paralysis of the legs and feet, bowel and bladder problems, learning problems, or hydrocephalus (too much fluid on the brain).

standard treatment
: Medical therapy that is widely accepted and used by most health care professionals as an appropriate treatment for a particular condition.

stroke
: A loss of blood flow to part of the brain. Strokes are caused by blood clots or broken blood vessels in the brain, and result in damage to a section of brain tissue. Symptoms include dizziness, numbness, weakness on one side of the body, and problems with talking or understanding language. The chance (risk) of stroke is increased by high blood pressure, older age, smoking, diabetes, high cholesterol, heart disease, a family history of stroke, and a build-up of fatty material inside the coronary arteries (atherosclerosis). See also NIH publication: Know Stroke. Know the Signs. Act in Time. http://www.ninds.nih.gov/disorders/stroke/knowstroke.htm

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom
: A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

treat
: To care for a patient with a disease by using medicine, surgery, or other approaches.

ulcerative colitis
: Chronic inflammation of the colon that causes ulcers to form in its lining. This condition is marked by abdominal pain, cramps, and loose discharges of pus, blood, and mucus from the bowel.

upper limit
: UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture
: USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin
: A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin B_{12}
: A group of chemical compounds that contain cobalt and are needed for certain chemical reactions in the body. Vitamin B_{12} is involved in maintaining healthy nerve cells and red blood cells. It is needed to make DNA (the genetic material in all cells), and is required for the metabolism (chemical changes that take place in the tissues to produce energy and the basic materials needed by the body) of carbohydrate, fat, and protein. Also called cobalamin. For more information see the NIH Office of Dietary Supplements Vitamin B_{12} fact sheet.

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